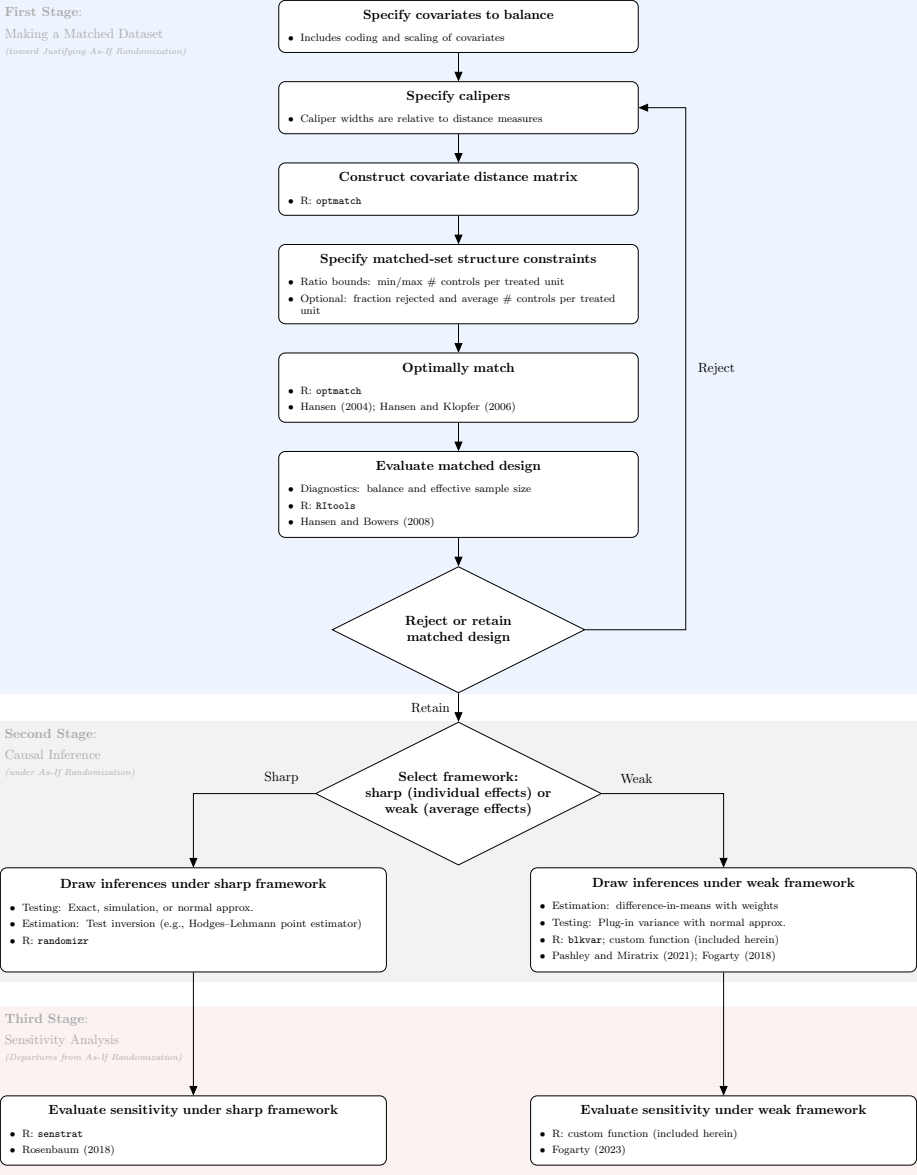

First Stage:
Making a Matched Dataset
(toward Justifying As-If Randomization)
Specify covariates to balance
Includes coding and scaling of covariates
Specify calipers
Caliper widths are relative to distance measures
Construct covariate distance matrix
R: optmatch
Specify matched-set structure constraints
Ratio bounds: min/max # controls per treated unit
Optional: fraction rejected and average # controls per treated unit
Optimally match
R: optmatch
Hansen (2004); Hansen and Klopfer (2006)
Evaluate matched design
Diagnostics: balance and effective sample size
R: RItools
Hansen and Bowers (2008)
Reject or retain matched design
Reject
Retain
Second Stage:
Causal Inference
(under As-If Randomization)
Select framework: sharp (individual effects) or weak (average effects)
Sharp
Weak
Draw inferences under sharp framework
Testing: Exact, simulation, or normal approx.
Estimation: Test inversion (e.g., Hodges–Lehmann point estimator)
R: randomizr
Draw inferences under weak framework
Estimation: difference-in-means with weights
Testing: Plug-in variance with normal approx.
R: blkvar; custom function (included herein)
Pashley and Miratrix (2021); Fogarty (2018)
Third Stage:
Sensitivity Analysis
(Departures from As-If Randomization)
Evaluate sensitivity under sharp framework
R: senstrat
Rosenbaum (2018)
Evaluate sensitivity under weak framework
R: custom function (included herein)
Fogarty (2023)